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High

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Medium

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Low

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Lesson 7: Percentages

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Two-Step Percentage Problems

In this lesson, you’ve learned that percents represent part of a whole. In many math problems and real-life situations, you must solve a percent equation, and then use the information to find another answer. The percent equation states that some percent of a whole equals the part.

Example 1

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Mario makes \$14 an hour. After one year on the job, he gets a 7% raise. What is Mario’s hourly wage after the raise?

Step 1
Set up the percent equation.
7% of \$14 is p .
 $.07 \times 14 = p$

Step 2
Solve for the unknown part.
 $.07 \times 14 = p$
 $.98 = p$

Step 3
Decide what operation is needed for the second step
Add the increase to the old hourly pay to find the new hourly pay.

Step 4
Solve.
 $\$0.98 + \$14.00 = \$14.98$
Mario now earns **\$14.98** per hour.

You can add a percent increase in one step by adding 100%. The raise was for 7%. That means the new hourly wage will be 107% of the current wage.

Example 2

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A jacket originally cost \$120. The store put the jacket on sale for 20% off. When the jacket still didn’t sell, the store reduced the price another 10%. What is the current sale price of the jacket?

Step 1
Find the amount of the first **discount**.
20% of \$120 is d .
 $.20 \times \$120 = \24
The first discount is \$24.

Step 2
Subtract the first discount.
 $\$120 - \$24 = \$96$
\$96 is the sale price after the first discount.

Step 3
Find the amount of the second discount.
10% of \$96 is d .
 $.10 \times \$96 = \9.60
The second discount is \$9.60

Step 4
Subtract the second discount.
 $\$96 - \$9.60 = \$86.40$
The current price of the jacket is **\$86.40**.

After two discounts, the price of the jacket was reduced $\$24 + \9.60 , for a total discount of \$33.60. This is not the same as taking a 30% discount once. A 30% discount takes \$36 off the price: $\$120 \times .30 = \36 . Always work the steps of a percent problem separately and in the order described in the problem.

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